
Refusal Reads the Harm Percept, Not the Moral Subspace

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Abstract

Alignment applied after pretraining is shallow in a measurable way: a refusal direction can be ablated with a rank-one edit. We ask what that refusal decision actually reads. Across four open-weight models (OLMo-3-7B, Qwen2.5-7B, Llama-3.1-8B, and the reasoning mixture-of-experts GPT-OSS-20B) we find that moral comprehension is pretraining-native: a rank-3 moral subspace crystallizes to a base-to-aligned cosine of 0.999 and post-training only rotates it once. The refusal gate, in contrast, is a fresh post-training construction (base-to-instruct cosine 0.155) written into a narrow 9-to-15 effective-dimensional control-token channel at the decision site, where the refusal decision is orthogonal to the moral-judgment decision. A nested interchange rank sweep on OLMo resolves what refusal reads: as the moral basis expands, judgment coupling climbs while refusal coupling saturates at the rank-1 harm level, so refusal reads the harm percept, a low-rank slice, not the broad subspace that judgment reads on the same patches. A four-model panel then separates two axes, what refusal reads (harm versus broad moral content) and how it commits (at the read layer, early, or reversibly). Refusal routes around the moral subspace; that is why it is easy to remove and hard to deepen.

1 Introduction

Alignment applied after pretraining is shallow in a specific, measurable sense: the behavior it installs can be removed cheaply. Reinforcement learning from human feedback (Ouyang et al., 2022), preference optimization (Rafailov et al., 2023), and constitutional methods (Bai et al., 2022) produce models that refuse harmful requests, yet the refusal behavior sits on a thin representational substrate. Arditi et al. (Arditi et al., 2024) show that refusal is mediated by a single direction in the residual stream and can be ablated with a rank-one edit; open tooling now removes it automatically (p-e-w, 2025). Safety training can also be circumvented from the inside: models trained to behave can conceal a triggered policy through the training itself (Hubinger et al., 2024), and can fake alignment when they infer they are monitored (Greenblatt et al., 2024). The common thread is that post-hoc alignment writes a shallow control on top of a deep model, and the depth of what the model *understands* is not the depth of what its refusal decision *uses*.

This paper asks a mechanistic question behind that gap: when a chat model decides to refuse, what does the decision read? A natural hypothesis, and the one a “deep alignment” program would hope for, is that refusal consults the model’s moral representations broadly, the same representations that let it judge scenarios as right or wrong. We find the opposite. Refusal reads the **harm percept**, a low-rank slice of moral content, and writes it into a narrow control-token bottleneck at the decision site. It does not read the broad moral subspace where comprehension lives. The two are nearly orthogonal, which is exactly why the refusal control is thin and removable while the comprehension underneath it is not.

We establish this across four open-weight models (OLMo-3-7B-Instruct, Qwen2.5-7B, Llama-3.1-8B, and the reasoning mixture-of-experts GPT-OSS-20B) with a combination of geometric measurement

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and causal interchange. The argument runs in seven steps. Moral comprehension is pretraining-native and survives alignment: a rank-3 moral subspace crystallizes during pretraining to a base-to-aligned cosine of 0.999, and post-training rotates it once and then leaves it. The refusal gate, by contrast, is a fresh post-training construction (base-to-instruct cosine 0.155) that lives in a low-variance channel. The decision site itself is a 9-to-15 effective-dimensional control-token bottleneck on all four architectures, and at that site the refusal-decision direction is orthogonal to the moral-judgment-decision direction. A nested interchange rank sweep on OLMo resolves *what* refusal reads: as the moral basis expands, judgment coupling climbs while refusal coupling saturates at the rank-1 harm level. Refusal reads harm; judgment reads the subspace broadly; they are different reads of the same content. A cross-model panel then separates two axes, *what* refusal reads (harm versus broad moral content) and *how* it commits (at the read layer, early, or reversibly), with GPT-OSS as an existence proof that a deliberating model can read harm and still be talked out of a refusal.

The measurement discipline behind these claims, the positive-control ladders, the position-validity gates, and the depth-referenced verdicts, is documented separately (a companion methods note, in preparation); here we cite it where a number depends on it and otherwise report the findings. Throughout, every null carries its detection bar, and every quantitative adjective carries its number.

2 Models and instruments

2.1 Models

The panel is four open-weight chat models spanning three lineages, two scales, a mixture-of-experts design, and an explicitly deliberative reasoning model: OLMo-3-7B-Instruct (Team OLMo, 2025), Qwen2.5-7B (Qwen Team, 2025), Llama-3.1-8B (Llama Team, AI @ Meta, 2024), and GPT-OSS-20B (OpenAI, 2025), a 20B reasoning mixture-of-experts. OLMo-3 is the primary target for the causal work because Ai2 releases base and post-training checkpoints, so we can watch a representation form during pretraining and track it through supervised fine-tuning and reinforcement stages. The other three provide cross-lineage and cross-architecture generalization for the representational cells.

2.2 Directions and subspaces

Three objects carry the argument, each a direction or a low-rank subspace in the residual stream, extracted by mean-difference over labeled stimulus contrasts in the spirit of representation reading (Park et al., 2024, Zou et al., 2023).

The moral subspace V_{moral} . We build one direction per moral-content source by mean-difference between moral and neutral stimuli, from three datasets: Moral Stories, Understanding Fables, and ETHICS (Hendrycks et al., 2021), grounded in moral foundations theory (Graham et al., 2013, Haidt, 2012). The three source directions are distinguishable rather than collinear (cosine between the fables direction and the pooled moral direction 0.53, between the ETHICS direction and the pooled direction 0.36) and their orthonormalized span has effective rank 3. V_{moral} is richer but lower-dimensional than a six-foundation moral-foundations span (three source directions versus six; effective dimension 3 versus 4); its value is construct diversity, source distinguishability, and resistance to single-source contamination, not extra dimensions. We refer to it in prose as the moral subspace.

The refusal direction. Following Arditi et al. (Arditi et al., 2024), we extract a single refusal direction by mean-difference between activations on requests the model refuses and requests it complies with. On the base model we also extract a *proto-refusal* direction from the same contrast, to ask whether the aligned gate has a pretraining precursor.

The judgment-decision direction. At the chat decision site we extract a moral-judgment direction by mean-difference between activations that precede an approving versus a disapproving judgment, so that both the refusal decision and the judgment decision are defined at the same token and can be compared directly.

2.3 Measurement conventions

Directions are compared by cosine and by projection fraction against calibrated nulls; subspaces by restricted interchange transfer (defined where used). Concept erasure, where needed, uses LEACE (Belrose et al., 2023). Two conventions matter for the numbers below. First, every projection is read

against a positive-control moral-family band (the projection of held-one-out moral directions onto the span of the rest) and a covariance-matched null, so that “below the band” and “below the null” are stated, not “small”. Second, participation ratio is recorded at every measurement position; a position with participation ratio below 30 is flagged invalid for content projection-fraction tests, for reasons that become the subject of Section 5. Details of these gates, and the positive-control ladders that certify them, are in the companion methods note (in preparation); we cite it at the points where a verdict rests on the protocol rather than restate it here.

3 Moral comprehension is pretraining-native and survives alignment

The first fact is about what alignment does *not* build. Moral comprehension is present and stable before any post-training touches the model, and post-training reorients it rather than teaching it.

On OLMo-3, where base and post-training checkpoints are released, we track the moral subspace across 25 pretraining and post-training states. Comprehension is pretraining-native: a linear probe on the moral subspace reaches 100% accuracy at every state, effective dimension holds at 5, and cross-source transfer AUC is ≈ 1.0 throughout. The subspace also *crystallizes* during pretraining. The cosine between the direction at each checkpoint and the fully trained direction rises from 0.869 at step 1000 to 0.999 by the end of pretraining. Once formed, the subspace is not rebuilt by alignment. Supervised fine-tuning rotates it once, from a base-to-SFT cosine of 0.999 down to 0.757 (about 40 degrees), and then it holds: preference optimization leaves it at 0.757, and every reinforcement substep stays in the 0.757–0.759 band, with effective dimension 5 throughout. Post-training reorients moral comprehension by a single rigid rotation; it does not re-teach it and does not change its rank.

This is consistent with what the pretraining studies in this line find about how moral structure emerges. Moral content is learned early and compositionally rather than as a bag of words (Reblitz-Richardson, 2026a), and the moral foundations a model acquires integrate into one positively correlated subspace rather than splitting into the theory’s individualizing and binding clusters (mean off-diagonal cosine 0.232–0.274, effective dimension 5 at every layer, first principal component 0.379 of variance against 0.179 for a random baseline, and no significant recovery of the moral-foundations split under permutation, minimum $p = 0.32$) (Reblitz-Richardson, 2026b). The picture is a moral representation that forms in pretraining, is broad and low-rank, and is preserved through alignment.

The contrast that organizes the rest of the paper is with the refusal gate. Where the moral subspace crystallizes to a base-to-aligned cosine of 0.999, the refusal gate reaches only 0.155 from its pretraining precursor. Comprehension is deep and inherited; the refusal decision, as the next section shows, is a shallow, freshly built control. Figure 1 plots the two side by side.

4 The refusal gate is a fresh post-training construction

The refusal decision is not a moral-content-derived direction inherited from pretraining. It is built during post-training, it lives in a low-variance channel, and it sits below the moral-family band on every model tested.

It does not crystallize from a precursor. We measure the cosine between the base model’s proto-refusal direction and the aligned model’s refusal gate at the layer where the gate is defined. It is 0.155, well below the 0.50 threshold that the moral subspace clears on its way to 0.999. The number is the headline pairing of Figure 1: 0.999 for comprehension, 0.155 for the gate. The aligned refusal gate is substantially a post-training construction, not a re-pointing of something the base model already had.

It lives in a low-variance channel. Across residual dimensions ranked by variance, the wired instruct refusal gate sits at the bottom: its variance percentile is 0.0 (within the lowest decile), and all four positions carrying the GPT-OSS refusal signal are likewise in the lowest decile. The moral-subspace axes and the persona direction, by contrast, occupy ordinary-to-high variance. The base model’s proto-refusal is not narrow (percentile 37.4), so the narrowness is a property of the *installed* gate, not of the raw contrast. Refusal is wired into a spare channel that carries little of the model’s activation variance.

It projects below the moral-family band at every rung. The base proto-refusal projects 0.33 onto the base moral subspace (covariance-matched null q95 0.291; the persona reference sits at 0.51), and the aligned gate projects 0.14 onto the aligned moral subspace (null q95 0.26). Across a rank

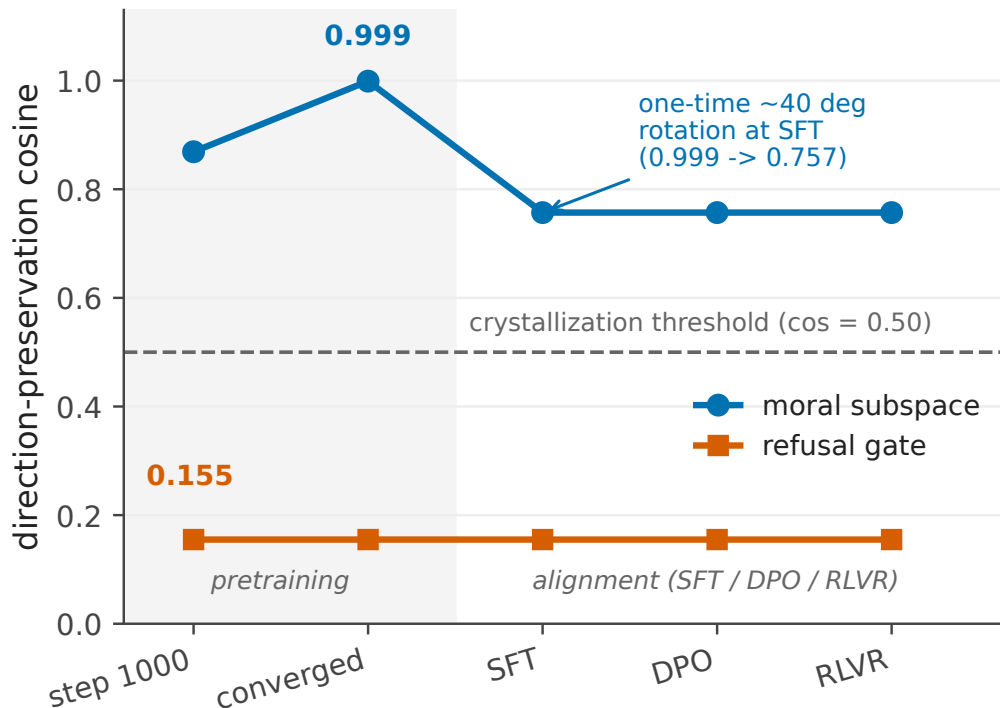


Figure 1: Comprehension crystallizes; the refusal gate does not. Left: the cosine between the OLMo-3 moral subspace at each training checkpoint and the fully trained direction rises from 0.869 at step 1000 to 0.999 during pretraining, then holds through post-training (supervised fine-tuning rotates it once to 0.757 and later stages leave it there). Right: the refusal gate’s cosine to its pretraining precursor is only 0.155, far below the 0.50 crystallization threshold. Moral comprehension is pretraining-native and inherited; the refusal decision is a fresh post-training construction.

sweep over the moral basis (one to three sources) refusal never clears the null-plus-margin bar. Read against a richer construction the story is the same: the aligned gate projects 0.144 onto the rank-3 moral subspace (null q95 0.266) and 0.155 onto the six-foundation moral-foundations span (null q95 0.252), both null. Figure 2 shows the calibrated ladder that turns these into verdicts: floor, matched null, measurement, and positive band on one axis, with refusal below the band.

These three readings converge. The refusal gate is a freshly built, low-variance post-training control, not a direction derived from the model’s moral content. That is the mechanism behind two otherwise separate facts: a rank-one edit can ablate refusal cleanly (Arditi et al., 2024, p-e-w, 2025), and refusal projects below the moral band. A control this thin and this orthogonal to comprehension is exactly what is cheap to remove. Consistent with this, on OLMo-3 the refusal direction projects only 0.10 of its norm into the moral subspace (mean $|\cos|$ 0.06), and ablating it drops refusal from 0.25 to 0.00 while leaving comprehension intact (base-to-fresh cosine 0.749, probe accuracy 1.0, effective dimension 5) and moral judgment essentially unchanged (0.73 versus 0.75).

5 The decision site is a control-token bottleneck

Where does the refusal decision live, and why is content-versus-decision orthogonality so easy to find there? The answer is a structural fact about the decision site that holds on every architecture we test.

The decision site is the control token where the chat template hands off to the model’s answer: the token before the assistant header on the instruct models, the end-of-prompt token in the reasoning model’s harmony format. This is where the refusal gate and the judgment direction are defined, because it is the last position the model reads before it commits to a reply. At that token the residual stream is a low-dimensional bottleneck. Its participation ratio is 14.7 on OLMo-3-7B-Instruct, 8.6 on

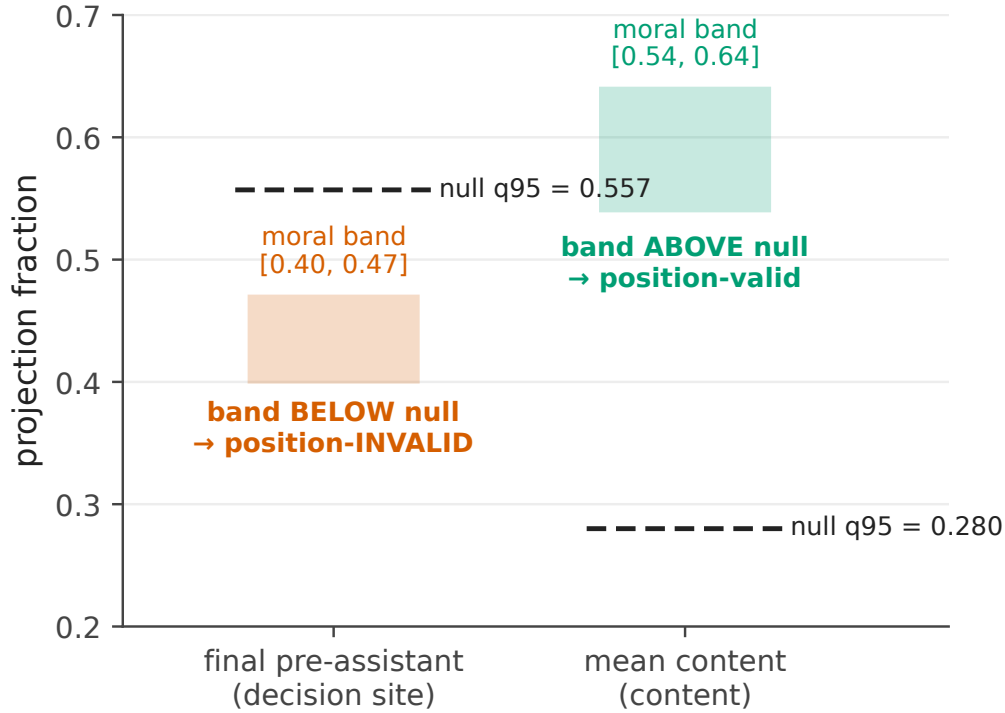


Figure 2: The calibrated projection ladder. For each tag, the refusal projection onto the moral subspace (marker) is plotted against a floor, a covariance-matched null, and the positive-control moral-family band (the projection of held-one-out moral directions onto the span of the rest). Refusal lands below the moral-family band on every model, including the in-trace peak on the reasoning models; even the program’s highest refusal projection is less moral-adjacent than a held-out moral direction. The ladder is the instrument that turns "small projection" into "below a stated bar".

Qwen2.5-7B, 10.2 on Llama-3.1-8B, and 12.79 on GPT-OSS-20B, a 9-to-15 effective-dimensional channel on all four models. Content positions at the same layers are full-rank-healthy by comparison (participation ratio above 40 on OLMo, above 33 on Qwen, above 35 on Llama). Figure 3 plots the four decision-site values against the position-validity gate. (The Llama value of record is 10.2, measured on the in-format ladder and directly comparable to OLMo’s 14.7 and Qwen’s 8.6; a separate decision-token harness reads 13.5 at a second position. Both are far below 30.)

This narrowness is the reason a projection-fraction instrument fails at the decision site, and it is also a substantive fact about where the decision lives. At the OLMo-3 decision token the positive-control moral band comes out at [0.40, 0.47], *below* the covariance-matched null of 0.557. A positive control below the null means the instrument has no discriminating power at that position: any direction, moral or not, projects onto a 15-slot channel at roughly the null level, so a low projection there cannot certify that a direction is absent from the subspace. Three independent estimates agree that the channel carries about 15 effective dimensions: $\sqrt{3/14.7} = 0.45$ as a closed-form projection expectation, the covariance null q95 of 0.557, and the pairwise-cosine null of 0.41–0.51. Because of this, participation ratio is a required field on every extracted direction, and any position below 30 is flagged invalid for content projection-fraction tests. The position-validity protocol and its positive-control ladder are documented in the companion methods note (in preparation).

One reconciling sentence is needed before the next section, because the bottleneck cuts two ways. It is position-invalid *for content projection-fraction tests* (the band-below-null tell), but position-valid *for decision-direction reads*: a cosine between two directions both defined at the decision token is immune to the projection null, and the GPT-OSS refusal projection is read at a decision channel that passes its own validity gate at participation ratio 12.79. So the bottleneck does not block the

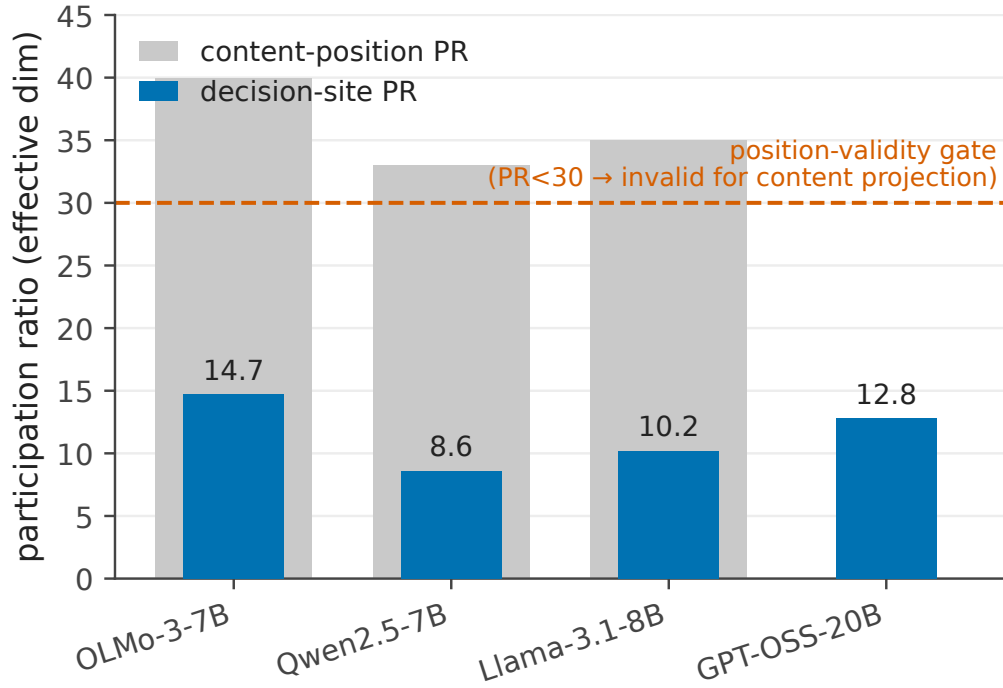


Figure 3: The decision-site participation ratio across four architectures: OLMo-3-7B-Instruct 14.7, Qwen2.5-7B 8.6, Llama-3.1-8B 10.2, and GPT-OSS-20B 12.79 (a 20B reasoning mixture-of-experts at its harmony decision token). All four fall below the participation-ratio 30 position-validity gate, while content positions at the same layers stay full-rank-healthy (above 40/33/35). The refusal decision lives in a 9-to-15 effective-dimensional control-token channel on every model tested.

comparison the next section makes; it blocks only the content-projection comparison, and it does so on all four architectures.

The structural consequence is the setup for the causal work. Content and the decision do not co-locate: moral content is readable at content positions (healthy participation ratio there) and unreadable at the decision channel (band-below-null), while the refusal and judgment directions live only at the decision channel. They never coexist at one valid position. Content-versus-decision orthogonality is therefore architecturally favored, not a discovered surprise, and any coupling between comprehension and the decision has to ride the attention heads that write into the bottleneck. That is a concrete anatomical target, and Section 7 pursues it.

6 Refusal is orthogonal to moral judgment at the decision

With both directions defined at the same valid decision token, the comparison is a single cosine per model, read against a calibrated null and the moral-family band. The refusal decision is orthogonal to the moral-judgment decision on every model.

On OLMo-3 the cosine between the refusal-decision and judgment-decision directions is 0.10, with no coupling detectable above $|\cos|$ 0.10 against a null q95 of 0.41: a margin of 0.35 below the minimum detectable effect (null q95 plus a 0.05 margin). On Qwen the cosine is 0.32 against a null q95 of 0.42 (margin 0.15), and on Llama it is 0.08 against a null q95 of 0.51 (margin 0.48). All three margins clear the detection bar; the refusal and judgment directions occupy different slots of the low-dimensional decision channel at a separation below even the random level for a channel that narrow. This reads as active separation, not a weak-instrument artifact: the same channel where a projection-fraction test has no power still supports a decision-direction cosine, and that cosine says the two directions are apart.

We state this as a bounded null, not a bare dissociation. On OLMo there is no coupling detectable above $|\cos| 0.10$ against a null $q95$ of 0.41; whatever small coupling exists is below that bar. This wording matters because the same measurement, the held-one-out moral-family band, also certifies that refusal is not merely far from judgment but *below the band of moral directions generally*. The bands are [0.537, 0.664] on the base model, [0.523, 0.637] on instruct, [0.537, 0.667] on the reasoning model, and [0.649, 0.764] on GPT-OSS, and every refusal point on every model lands below its band (the base band-minimum 95% confidence interval is [0.47, 0.53], and refusal sits under it). Figure 2 shows the same fact from the projection side.

The orthogonality is real but, on its own, under-determined. Geometric non-overlap does not by itself establish that refusal reads none of the moral content; it establishes only that the refusal-decision direction and the judgment-decision direction point in different places, and Section 5 already showed that content-versus-decision orthogonality is structurally favored at this site. To learn what refusal actually reads, and whether the moral subspace is causally inert for it or merely read through a narrow slice, requires a causal cell on the heads that write the decision channel. That is the next section.

7 What refusal reads: the harm percept

The causal core is on OLMo-3, where interchange patching (Meng et al., 2022) on the heads that write the decision channel resolves *what* the refusal decision reads. The answer is the harm percept: a rank-1 slice of the moral subspace, aligned with a harm direction (Zhao et al., 2025), and not the broad subspace that moral judgment reads on the same patches. Moral directions here are causal and foundation-specific to begin with, a preliminary we established with foundation-wise ablation whose specificity strengthens with depth; the OLMo-3 interchange cells below sharpen that into a rank sweep that says *which* moral content refusal uses.

The write is distributed. Refusal is written into the ~13-dimensional decision channel by a set of heads, led by layer 16 head 23 but not carried by it. That head writes +0.742 onto the refusal direction (channel-matched specificity +0.756) and alone accounts for 11.6% of the total specificity; the writers span layers 11–16, with layer 15 head 15 the sole anti-refusal writer (−0.130). Cumulative channel-matched specificity reaches 44% at the top ten heads and needs about 62 heads for 80%. Attention is not the whole story: multilayer perceptrons contribute 38% of the decision-site write (write fraction 0.384). The per-head attribution is certified by an exact normalization fold that brings the reconstruction from 3.05 to 0.9999 (companion methods note, in preparation). None of the top ten writers is a clean harm-copy head: all are labeled neither-moral-nor-harm, with a moral-subspace fraction of 0.15–0.28 and comparable harm loading, split between instruction-attenders and content-attenders. Refusal is written broadly, not routed through one moral head.

The interchange is specific to the moral subspace, then specific to harm inside it. Using request-twins (matched requests carrying opposite judgment outcomes, $n = 23$), we patch the decision channel and read the induced change in the refusal and judgment projections. The full patch moves refusal −0.0833; a patch restricted to the moral subspace moves it −0.0282; its complement moves it −0.0636; a harm-rank-1 patch moves it −0.0261; a random rank-3 patch moves it −0.0005 (refusal minimum detectable effect 0.0238). The moral subspace is a *specific* substrate: restricting to it moves refusal more than a random rank-3 patch does ($\Delta = 0.031$, paired 95% CI [0.020, 0.043], excludes 0). But almost all of that specific effect is the harm slice. The harm-restricted patch (−0.0261) nearly equals the full moral-subspace patch (−0.0282), and the harm-partial patch (the moral subspace with the harm direction projected out) still moves refusal −0.0133 (95% CI [−0.023, −0.005], excludes 0), about half. So refusal is harm-dominant with a small, resolvable residual non-harm moral read (the harm direction captures a fraction 0.46 of the moral subspace).

The rank sweep is decisive. For a readout r (the refusal or the judgment projection), define the restricted-transfer coefficient $R_r(k)$ as the fraction of the full interchange effect on r that is reproduced when the patch is confined to the top- k directions of the moral subspace. As k grows over $\{1, 3, 8, 16\}$, judgment transfer climbs $0.05 \rightarrow 0.46 \rightarrow 0.59 \rightarrow 0.66$ while refusal transfer saturates $0.01 \rightarrow 0.31 \rightarrow 0.26 \rightarrow 0.27$ at the harm-rank-1 level (harm-rank-1 transfer 0.31), with a random-direction null near zero at every rank and per-rank purity 0.97–0.99. Expanding the moral basis beyond harm buys more judgment coupling and no more refusal coupling. This is the central finding, and Figure 4 plots it: refusal reads the harm percept and stops; judgment keeps reading as the subspace widens. About 73% of refusal’s causal twin-difference input lies outside the rank-16 moral

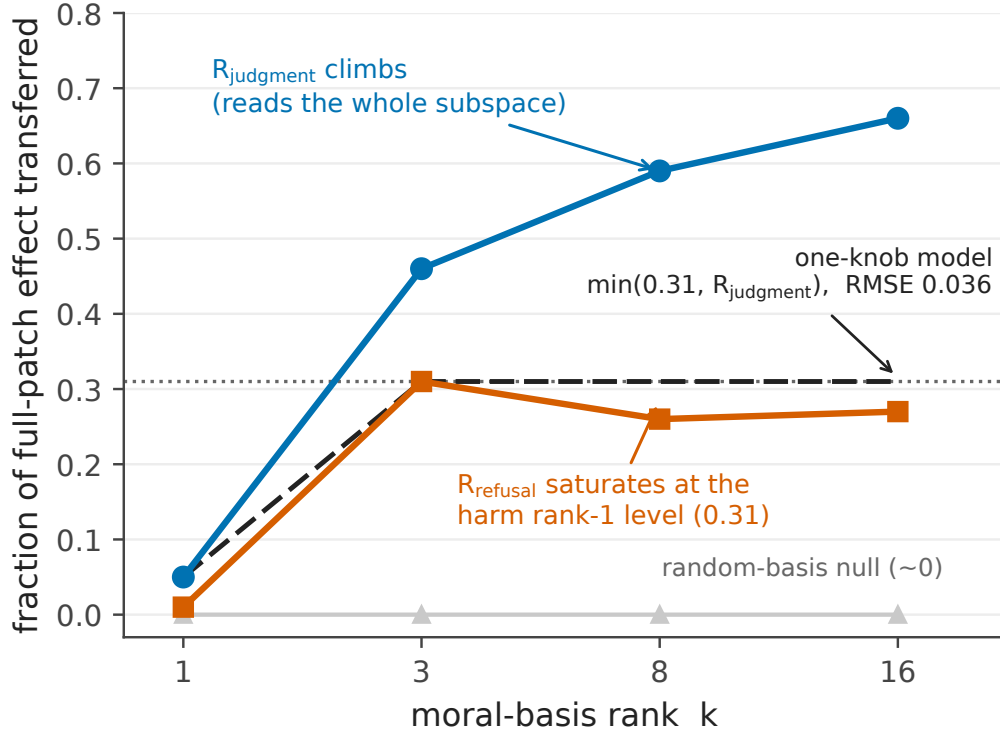


Figure 4: The nested rank sweep on OLMo-3, the paper’s central causal result. As the moral basis expands ($k \in \{1, 3, 8, 16\}$), judgment transfer $R_{\text{judgment}}(k)$ climbs $0.05 \rightarrow 0.46 \rightarrow 0.59 \rightarrow 0.66$ (open markers) while refusal transfer $R_{\text{refusal}}(k)$ saturates $0.01 \rightarrow 0.31 \rightarrow 0.26 \rightarrow 0.27$ at the harm-rank-1 level (filled markers); a random-direction null is near zero throughout. The dashed curve is the one-knob fit $R_{\text{refusal}}(k) \approx \min(\text{harm ceiling} \approx 0.31, R_{\text{judgment}}(k))$, RMSE 0.036 on the plateau. Refusal reads the harm percept and stops; judgment reads the subspace broadly on the same patches.

basis (69% already at the rank-3 peak). Judgment reads the subspace broadly *on the same patches*, which is the within-model proof that the content is there to be read; refusal simply does not read it.

One free parameter fits the sweep. The refusal curve is the judgment curve clipped at a harm ceiling: $R_{\text{refusal}}(k) \approx \min(\text{harm ceiling}, R_{\text{judgment}}(k))$, with the ceiling ≈ 0.31 . This one-knob model fits the plateau ($k \geq 3$) at RMSE 0.036 (residual -0.002 at $k = 3$), while harm-amplitude alternatives miss by 0.10–0.24. The one place it breaks is rank 1, where it over-predicts (measured 0.013 versus predicted 0.052): the highest-variance contrast component, the most harm-aligned single direction (variance purity 0.974, cosine 0.35 to harm), is causally inert, moving neither readout at rank 1 ($R_{\text{refusal}}(1) = 0.01$, $R_{\text{judgment}}(1) = 0.05$). Variance is not causal relevance; the harm read is a rank-1 causal object that is not the rank-1 variance object.

Behaviorally, this harm-keyed, saturating read is coherent with OLMo-3 being a weak intent-refuser. On intent-harmful requests its refusal reaches only about 17% at top severity (violating items 0/0.17/0/0.17/0.17 across a severity ladder, benign items 0). The operating band is nearly empty, which is a model property, weak coupling between intent severity and refusal, that is exactly what a harm-surface-keyed gate predicts, and it is the reason the cross-model commitment axis in Section 8 is measured on Llama and GPT-OSS rather than on OLMo alone.

8 Across families: what refusal reads and how it commits

The OLMo result is one model. A four-model panel shows that the harm-reading picture generalizes but that refusal decisions differ along two separable axes: *what* they read (harm versus broad moral content) and *how* they commit (at the read layer, early, or reversibly).

The harm direction is a real, causal, cross-model object. Before separating the axes, the harm percept has to be more than an OLMo artifact, and it is. Harmfulness and refusal are separately encoded at the instruction token, with the harmfulness read discriminating cleanly (d-prime 5.01 on GPT-OSS) and near-orthogonal to the downstream refusal read (cosine 0.16), extending Zhao et al. (Zhao et al., 2025) to deliberative reasoning models. The harm direction is largely outside the moral subspace, with an in-subspace fraction of 0.18 on GPT-OSS and 0.11 on reasoning distills, 3.0–3.9 times the $\sqrt{k/d} \approx 0.04$ chance floor, so about 85% of it lies outside the moral foundations, the same “reads a harm sliver” object the OLMo rank sweep formalizes. And it is causal: reply-inversion steering along the harm direction flips model judgments (Qwen2.5-14B-Instruct shift +17.4 flips 33% of replies, Llama-3.1-8B-Instruct +3.0 flips 23%), where an earlier raw diff-of-means null was a magnitude artifact (Zhao et al., 2025).

8.1 Llama reads broad and commits early

Llama-3.1’s anatomy is OLMo-like: pre-norm reconstruction 1.0008 (no fold needed), a clean low-dimensional decision channel (participation ratio 13.5, null 0.148 moving to 0.114 under standardization), a distributed write with a 30% multilayer-perceptron share, and all top writers labeled neither-moral-nor-harm. But Llama refuses on intent (baseline refusal 9/10, against OLMo’s ~17%), so its refusal cell is measurable where OLMo’s is empty, and it reads differently.

At matched depth (layer 12, chosen below) Llama reads the moral subspace *broadly*: refusal transfer 0.85 is essentially equal to judgment transfer 0.79, the gap that stays open on OLMo closes on Llama, and a harm-rank-1 restriction recovers only 0.59. The reads-broad verdict survives a harm-coextensive alternative at rank 1: a single harm cue spans only 3.6% of the moral basis that drives Llama’s refusal, so the transfer grows into moral directions the harm axis does not point along (the rank-2/4 severity-ladder version of this control is a stated extraction rider on contrasts not yet saved). Llama reads broad moral content, not just harm.

The commitment axis is why the matched-depth qualifier is load-bearing. Llama’s refusal is directionally asymmetric at the decision boundary (36 micro-graded twins): adding harmful content moves refusal coherently (+0.142, 95% CI [+0.086, +0.212], sign fraction 0.81), but removing harmful content does not (disengage −0.014, 95% CI [−0.084, +0.052], sign fraction 0.51, incoherent). A patch-layer sweep names the mechanism: Llama’s disengage is coherent below the read layer (−0.12/ −0.11/ −0.20 at layers 8/12/14, confidence intervals excluding 0; the depth-matched full cell at layer 12 reads −0.57) but incoherent at the read layer 16 (−0.014). OLMo’s disengage, by contrast, is coherent at the read layer (−0.62). Llama commits *early*, crystallizing its refusal before the decision site; OLMo commits at or after the read layer.

This resolves a robustness anomaly in the same panel. Llama’s refusal is entangled with moral judgment where the other models’ is not: at the best ablation layer, removing refusal drops judgment accuracy from 0.75 to 0.604, a -21σ outlier against matched-random ablations (0.747 ± 0.007) and dose-dependent (Spearman 1.0). Early commitment of a broad moral read is the mechanism: because Llama reads broad moral content and commits before the decision site, ablating its refusal reaches into the moral read in a way OLMo’s harm-keyed late-committing gate does not. The cross-model asymmetry that first looked like a third property is instead a consequence of early commitment. Read at the layer where each model commits, the naive asymmetry statistic is +0.82; read at matched depth (layer 12), it collapses to −0.28 on Llama against −0.54 on OLMo, a difference of +0.26. Figure 5 shows the collapse, the depth-referenced verdict that separates a genuine difference from a measurement taken past the commitment layer.

8.2 GPT-OSS reads harm and is reversible

GPT-OSS reads harm, but correlationally rather than by interchange. Its refusal direction is harm-loaded at both positions it carries signal: at the prompt (the instruction token) the standardized cosine to the harm direction is 0.977 against 0.001 for the harm-orthogonal moral subspace (near-purely harm), and in-trace it stays harm-dominant but attenuates (0.49 versus 0.13). This is a prompt-to-trace consistent harm read, and it is why the in-trace refusal projection landed below the moral-family band earlier. It is a projection result, not a patching result, so the reads-harm placement for GPT-OSS is correlational; the causal interchange version is held.

Table 1: What refusal reads $\times$ how it commits, across three model families. Rows are the models with a resolved commitment reading; columns are the two axes. OLMo’s read is by interchange, Llama’s by interchange at matched depth, GPT-OSS’s by projection (correlational).

Model	What refusal reads	How it commits
OLMo-3-7B	Harm percept (transfer saturates at the harm-rank-1 level, ceiling 0.31; judgment reads broadly)	At / after the read layer (disengage coherent at the read layer, -0.62)
Llama-3.1-8B	Broad moral content (refusal transfer $0.85 \approx$ judgment 0.79 at matched depth, gap closes)	Early (disengage coherent below layer 15, incoherent at the read layer 16)
GPT-OSS-20B	Harm (correlational: prompt cosine 0.977 to harm vs 0.001 orthogonal; causal test held)	Reversible reader (engage 7/7, disengage 6/10, monotone projection)

What GPT-OSS adds is the commitment axis at its most informative extreme: it is a *reversible reader*. An inculcating-analysis prefill flips unsaturated benign requests to refuse 7 out of 7 (Wilson 95% [0.65, 1.0]), so the decision is not fixed before the trace, deliberation is consequential. And in the other direction, a graded exculpatory prefill flips ceiling-refusing violating items to comply 6 out of 10, with the decision-channel refusal projection moving monotonically toward comply in all 10 items (projection-moved fraction 1.0, monotone fraction 1.0). Figure 6 shows the graded panel. GPT-OSS reverses in both directions; its refusal is a read that deliberation can re-argue, the clean contrast to Llama’s early commitment.

8.3 The two-axis result

Table 1 states the measured result: refusal reads harm on OLMo and GPT-OSS and broad moral content on Llama, and it commits at the read layer on OLMo, early on Llama, and reversibly on GPT-OSS. This table is the empirical claim, and it stands.

Its *interpretation* is a hypothesis, not an $n = 3$ result. The three points are ordinally consistent with a single underlying knob: the models whose refusal reads a low-rank harm slice (OLMo and GPT-OSS, roughly rank 1) are the ones that commit late or reversibly, and the model whose refusal reads broadly (Llama, roughly rank 8) is the one that commits early. This licenses “the effective dimensionality of the refusal read predicts its reversibility” as a falsifiable follow-on hypothesis. It does not confirm it. The read-and-commit pairing is architecture-confounded at three points: the models differ in lineage, scale, tokenizer, and reasoning-versus-instruct training all at once, so a dimensionality account and a lineage account fit the same table equally well. Deconfounding requires varying one axis at a time, for example a deliberation-trained variant of a single base model, or a lineage-matched scale sweep. We state the hypothesis to be tested, not a mechanism established.

9 Discussion

A mechanism for shallow alignment. The pieces compose into a concrete account of why post-hoc alignment is shallow. Moral comprehension is deep and inherited: a broad, low-rank moral subspace forms during pretraining and survives alignment as a single rigid rotation (Section 3). The refusal decision is a thin control built on top of it: a fresh post-training gate (Section 4) that lives in a narrow control-token channel (Section 5) and reads only the harm percept, a rank-1 slice of the moral subspace nearly orthogonal to the bulk (Section 7). The compliance wrapper reads a harm cue over a narrow bus; the model’s moral understanding sits mostly off that bus. This is not that comprehension is causally absent, the same patches that barely move refusal move judgment across the whole subspace, but that refusal does not consult it. Shallow alignment is shallow because the refusal decision reads a small, separable feature rather than the model’s moral representation.

Why refusal is easy to remove. The account predicts the removability that motivated this work. A control that occupies a low-variance channel and reads a rank-1 harm slice is a small target: a rank-one edit that cancels the refusal direction leaves the moral subspace, which the refusal decision

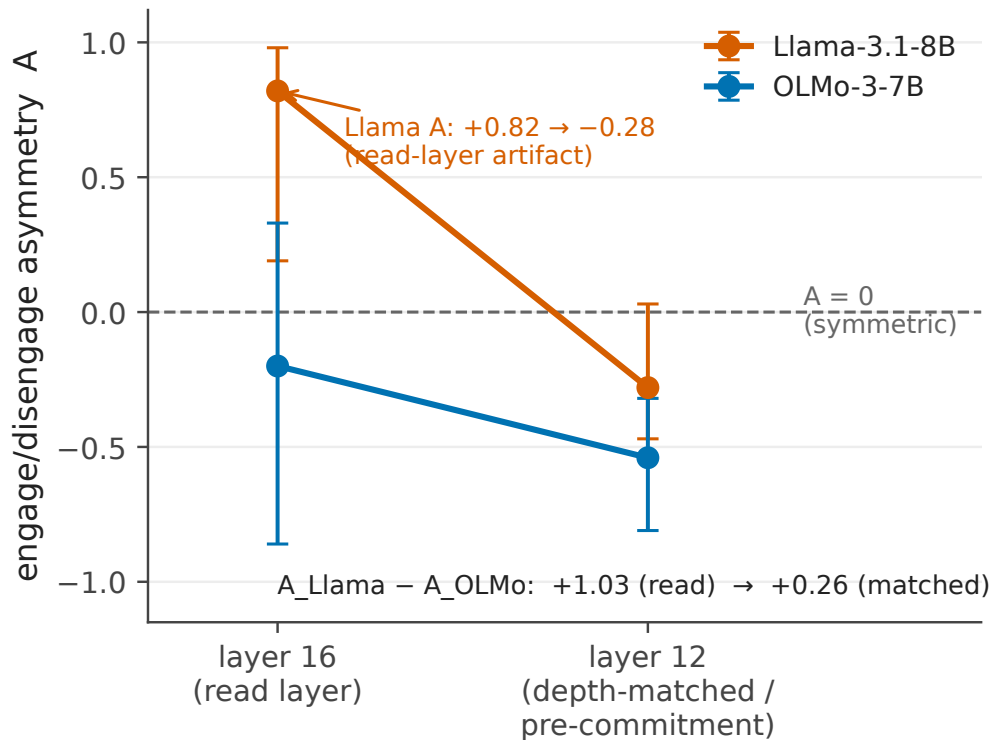


Figure 5: Depth-referenced verdicts, the Llama-versus-OLMo asymmetry. Read at each model’s own read layer, Llama’s refusal asymmetry statistic is $+0.82$, which reads as a third property (a hard latch). Read at matched depth (layer 12), it collapses to -0.28 against OLMo’s -0.54 , a difference of $+0.26$. The read-layer value was a post-commitment artifact: Llama commits early, so a measurement at its read layer is taken past the layer where the decision was already fixed. The asymmetry is a consequence of early commitment, not a separate axis.

was not reading, untouched (Arditi et al., 2024), which is why automated censorship removal succeeds without degrading the model (p-e-w, 2025). Removability is not a surprising fragility; it is what a harm-keyed gate over a narrow bus looks like from the outside. The same geometry explains why safety behavior can be trained shallowly enough to be concealed (Hubinger et al., 2024) or faked under monitoring (Greenblatt et al., 2024): the decision that governs the behavior is not wired into the representation that would have to change for the behavior to change deeply.

A forward target. If shallow alignment is refusal reading a harm sliver, deep alignment is refusal reading more of the moral subspace. The anatomy makes the target precise: the heads that write the decision channel (Section 7) currently transfer the harm rank-1 direction and saturate there, while judgment transfer keeps climbing as the basis widens. The intervention is to make those writing heads read the directions judgment already reads, so that refusal transfer follows the judgment curve instead of clipping at the harm ceiling. The small non-harm residual the interchange already resolves (the harm-partialed patch still moves refusal, about half the harm effect) is the toehold: it is the one place refusal demonstrably reads moral content beyond harm, so a rank-2 non-harm sliver, not the broad subspace, is where any deepening would begin. Whether widening the read also deepens the behavior, and at what cost to the model, is the question the two-axis panel raises and does not answer.

Deliberation can be load-bearing and reversible. GPT-OSS is an existence proof that the harm-keyed reflexive gate is not the only design point. Its refusal reads harm like OLMo’s, but its commitment is reversible: a graded exculpatory argument flips a ceiling refusal to compliance, and an inculcating argument flips a benign request to refusal (Section 8.2). A deliberating model can hold its refusal open to argument, which is a different, and in some respects more auditable, safety property than a fixed reflexive gate, because the decision is exposed to and moved by explicit reasoning rather than settled before the trace begins. It also has a failure mode the reflexive gate does not: a refusal

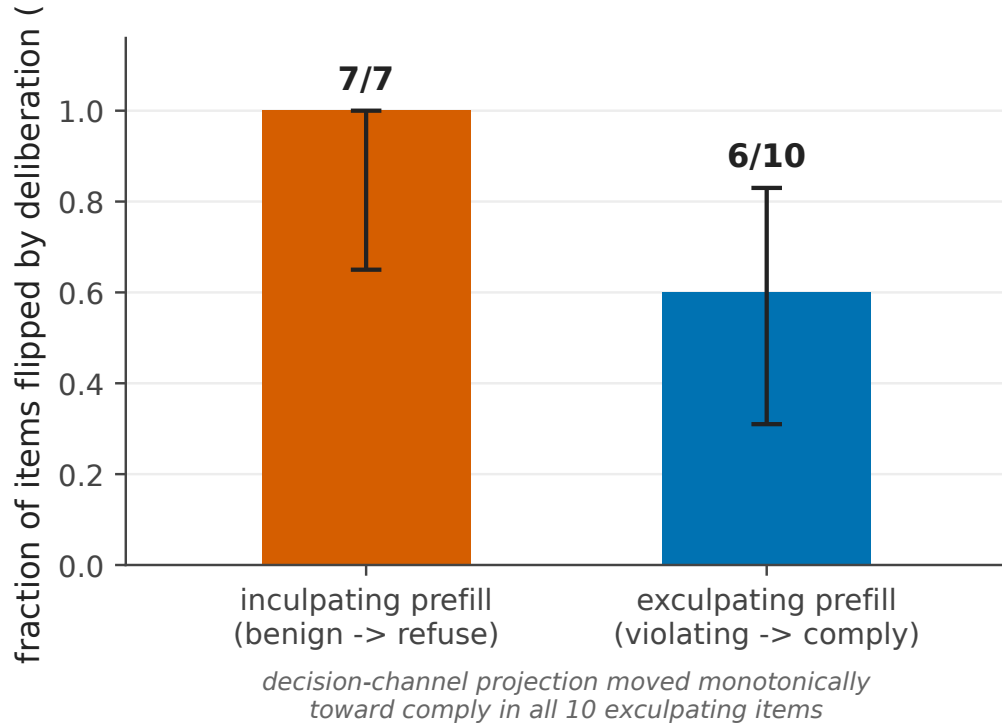


Figure 6: GPT-OSS is a reversible reader. A graded exculpatory-analysis prefill (increasing strength, left to right) flips ceiling-refusing violating items toward compliance, 6 of 10 flipping behaviorally, while the decision-channel refusal projection moves monotonically toward comply in all 10 items (monotone fraction 1.0). In the other direction an inculcating prefill flips benign requests to refuse 7 of 7. Deliberation is consequential and reversible in both directions, the clean contrast to Llama’s early commitment.

that can be argued down can be argued down by an adversary’s prefill. The two-axis panel makes the design choice explicit rather than settling it.

10 Limitations

The panel is three points and confounded. The two-axis table (Table 1) is a measured result, but its interpretation as a dimensionality-to-reversibility law is a hypothesis on three models that differ in lineage, scale, tokenizer, and reasoning-versus-instruct training simultaneously. A one-axis account (the effective dimensionality of the refusal read predicts reversibility) and a lineage account fit the same three points equally well. We state the hypothesis; we do not claim the mechanism. Deconfounding needs one axis varied at a time, for example a deliberation-trained variant of a single base model, or a lineage-matched scale sweep.

GPT-OSS reads-harm is correlational. GPT-OSS is placed on the harm-reading axis by projection (its refusal direction is harm-loaded at the prompt, cosine 0.977 to harm versus 0.001 to the harm-orthogonal moral subspace), not by interchange. The causal version, the nested rank sweep that resolves the OLMo verdict, is held for GPT-OSS. So the OLMo harm-reading verdict is causal and the GPT-OSS one is correlational, and the paper marks the two differently.

Readout versus behavior scope differs by cell. Some cells read internal directions (the rank sweep, the decision-channel projections) and some read behavior (the 7/7 engage flip, the 6/10 disengage flip, the ~17% OLMo refusal rate). These are different outcome variables, and a result on one does not automatically transfer to the other. Where a cell is a projection read we say so; where it is a behavioral flip we say so; we do not silently promote a projection movement to a behavior change.

The prefill-last-token projection caveat. The GPT-OSS reversibility result is primary on the behavioral flip (6/10 violating items flipped to comply) and corroborated by the decision-channel projection moving monotonically toward comply in all 10 items. That projection is read at the last token of the prefill, a position whose activation carries prefill-specific content; the monotone projection is corroboration for the behavioral flip, not an independent causal claim.

Stimulus-composition covariates across model bands. The moral-family bands and null values are computed per model on its own activation sample, and the stimulus sets that define the positive-control bands are not identical in composition across models. Cross-model comparisons of absolute projection values therefore carry a stimulus-composition covariate; the within-model verdicts (below its own band, below its own null) do not, and those are the ones we report as findings.

OLMo’s weak behavioral coupling. OLMo-3’s refusal reaches only about 17% at top intent severity, so its behavioral operating band for intent-graded refusal is nearly empty. This is coherent with the harm-surface-keyed read (a weak intent-refuser is what a harm-keyed gate predicts) and we report it as a model property, but it means the OLMo commitment-axis and severity-graded behavioral cells are measured on Llama and GPT-OSS, where refusal tracks intent, rather than on the model that carries the causal rank sweep.

11 Conclusion

Refusal reads the harm percept, not the moral subspace. In open-weight chat models the moral representation is deep and inherited, a broad low-rank subspace that forms in pretraining (crystallizing to a base-to-aligned cosine of 0.999) and survives alignment as a single rotation. The refusal decision built on top of it is shallow by construction: a fresh post-training gate (base-to-instruct cosine 0.155) in a narrow control-token channel, reading a rank-1 harm slice that a nested interchange sweep on OLMo shows saturating while judgment coupling keeps climbing. A four-model panel separates *what* refusal reads (harm versus broad moral content) from *how* it commits (at the read layer, early, or reversibly), with GPT-OSS as an existence proof that a deliberating model can read harm and still be argued out of a refusal. The positive claim is sharp and actionable: alignment is shallow because the refusal decision reads a small separable feature rather than the model’s moral understanding, and the same anatomy that explains why refusal is easy to remove also names where deeper alignment would have to write, into the directions judgment already reads.

Safety scope. This work characterizes the refusal geometry of released open-weight models, in order to explain a known property (that refusal is cheap to remove) and to locate where a future intervention would act. It reports what these models do; it does not build or optimize a method for removing refusal, and the forward target it names, widening what the writing heads read, is a direction for deeper alignment, not for defeating it. The interchange, ablation, and prefill cells are measurements on the models’ own forward passes, run to understand the decision, not to weaken it.

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